# Hands-on Lab: Working with Facts and Dimension Tables

**Estimated time needed: 30 minutes**

## Purpose of the lab:

The lab is designed to guide you through the process of designing a data warehouse for a cloud service provider. It focuses on using billing data provided in a CSV file to create a star schema, including the design of fact and dimension tables. This schema will support complex queries related to billing, such as average billing per customer, billing by country, industry, and category, as well as trends over time.

## Benefits of Learning the Lab:

By completing this lab, you will acquire practical skills in organizing and analyzing large datasets using data warehousing techniques. These skills are essential for making informed business decisions, optimizing data retrieval, and enhancing the understanding of data relationships. This knowledge is particularly beneficial in real-world scenarios, such as analyzing cloud billing data, where it can lead to more efficient data management and insightful analyses.

## Objectives

In this lab you will:

- Study the schema of the given csv file
- Design the fact tables
- Design the dimension tables

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- Create a star schema using the fact and dimension tables

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# About Skills Network Cloud IDE

Skills Network Cloud IDE (based on Theia and Docker) provides an environment for hands on labs for course and project related labs. Theia is an open source IDE (Integrated Development Environment), that can be run on desktop or on the cloud. To complete this lab, we will be using the Cloud IDE based on Theia running in a Docker container.

## Important Notice about this lab environment

Please be aware that sessions for this lab environment are not persistent. A new environment is created for you every time you connect to this lab. Any data you may have saved in an earlier session will get lost. To avoid losing your data, please plan to complete these labs in a single session.

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# Exercise 1: Study the schema of the given csv file

In this lab, we will design a data warehouse for a cloud service provider.

The cloud service provider has given us their billing data in the csv file `cloud-billing-dataset.csv`. This file contains the billing data for the past decade.

Here are the field wise details of the billing data.

| Field Name | Details |
|---|---|
| customerid | Id of the customer |
| category | Category of the customer. Example: Individual or Company |
| country | Country of the customer |
| industry | Which domain/industry the customer belongs to. Example: Legal, Engineering |
| month | The billed month, stored as YYYY-MM. Example: 2009-01 refers to the month January in the year 2009 |
| billedamount | Amount charged by the cloud services provided for that month in USD |

We need to design a data warehouse that can support the queries listed below:

- average billing per customer
- billing by country
- top 10 customers
- top 10 countries
- billing by industry
- billing by category
- billing by year
- billing by month
- billing by quarter
- average billing per industry per month

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- average billing per industry per quarter
- average billing per country per quarter
- average billing per country per industry per quarter

Here are five rows picked at random from the csv file.

| customerid | category | country | industry | month | billedamount |
|---|---|---|---|---|---|
| 1 | Individual | Indonesia | Engineering | 2009-1 | 5060 |
| 614 | Individual | United States | Product Management | 2009-1 | 9638 |
| 615 | Individual | China | Services | 2009-1 | 11573 |
| 616 | Individual | Russia | Accounting | 2009-1 | 18697 |
| 617 | Individual | Chile | Business Development | 2009-1 | 944 |

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# Exercise 2: Design the fact tables

The fact in this data is the bill which is generated monthly.

The fields `customerid` and `billedamount` are the important fields in the fact table.

We also need a way to identify the additional customer information, other than the id, and date information. So we need fields that refer to the customer and date information in other tables.

The final fact table for the bill would look like this:

| Field Name | Details |
|---|---|
| billid | Primary key - Unique identifier for every bill |
| customerid | Foreign Key - Id of the customer |
| monthid | Foreign Key - Id of the month. We can resolve the billed month info using this |
| billedamount | Amount charged by the cloud services provided for that month in USD |

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# Exercise 3: Design the dimension tables

There are two dimensions to our fact(monthly bill).

1. Customer information
2. Date information

Let us organize all the fields that give information about the customer into a dimension table.

| Field Name | Details |
|---|---|
| customerid | Primary Key - Id of the customer |
| category | Category of the customer. Example: Individual or Company |
| country | Country of the customer |
| industry | Which domain/industry the customer belongs to. Example: Legal, Engineering |

Let us organize or derive all the fields that give information about the date of the bill.

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| Field Name | Details |
| --- | --- |
| monthid | Primary Key - Id of the month |
| year | Year derived from the month field of the original data. Example: 2010 |
| month | Month number derived from the month field of the original data. Example: 1, 2, 3 |
| monthname | Month name derived from the month field of the original data. Example: March |
| quarter | Quarter number derived from the month field of the original data. Example: 1, 2, 3, 4 |
| quartername | Quarter name derived from the month field of the original data. Example: Q1, Q2, Q3, Q4 |

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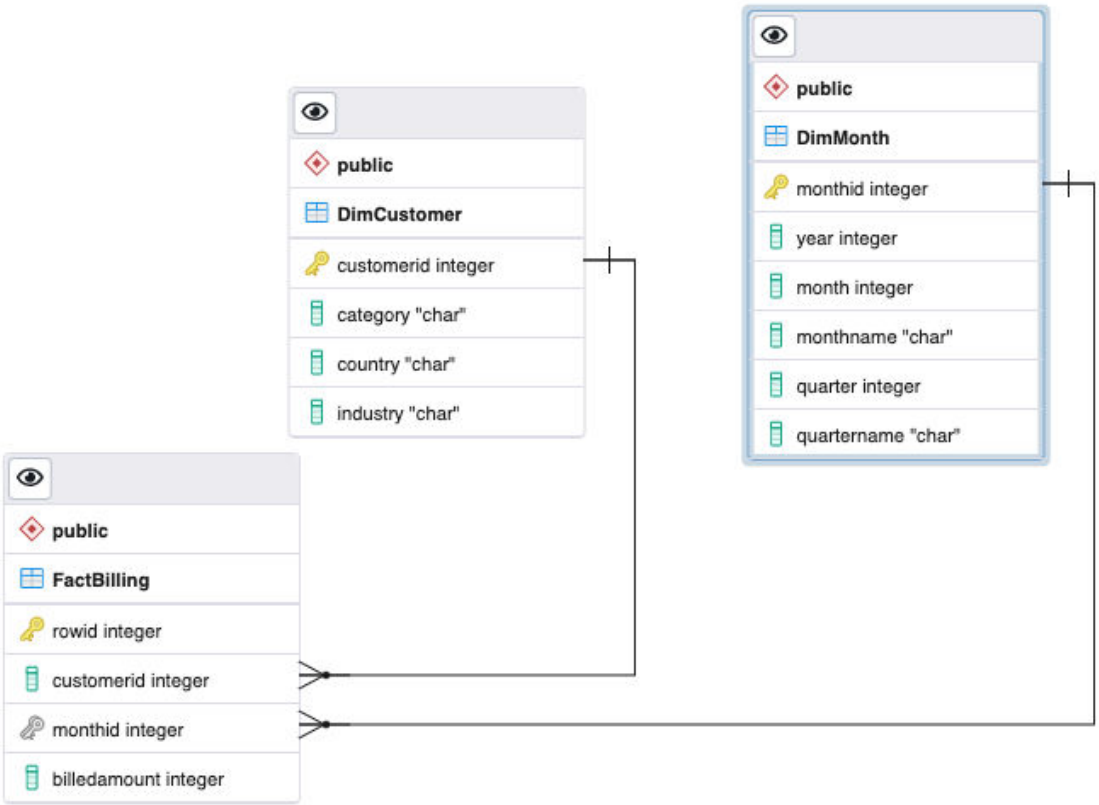
# Exercise 4: Create a star schema using the fact and dimension tables

Based on the previous two exercises, we have now arrived at 3 tables, we can name them as in the table below.

| Table Name | Type | Details |
|---|---|---|
| FactBilling | Fact | This table contains the billing amount, and the foreign keys to customer and month data |
| DimCustomer | Dimension | This table contains all the information related the customer |
| DimMonth | Dimension | This table contains all the information related the month of billing |

When we arrange the above tables in Star Schema style, we get a table strucutre that looks likes the one in the image below.

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The image shows the fact and dimension tables along with the relaionships between them.

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# Exercise 5: Create the schema on the data warehouse

Step 1: Start the postgresql server.

Start the PostgreSQL server by clicking the command below:

Open and Start PostgreSQL in IDE

Step 2: Create the database on the data warehouse.

Using the createdb command of the PostgreSQL server, we can directly create the database from the terminal.

Firstly, run the command below to set your PostgreSQL password for authentication. Replace with your actual PostgreSQL password, and then execute the command:

bash

```bash
export PGPASSWORD=<your_password>
```

Run

Now, run the command below to create a database named billingDW.

bash

```bash
createdb -h postgres -U postgres -p 5432 billingDW
```

Run

In the above command

- -h mentions that the database server is accessible using the hostname "postgres"
- -U mentions that we are using the user name postgres to log into the database
- -p mentions that the database server is running on port number 5432

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```
theia@theiadocker-rsannareddy:/home/project$ createdb -h localhost -U postgres -p 5432 billingDW
theia@theiadocker-rsannareddy:/home/project$
```

Step 3: Download the schema .sql file.

The commands to create the schema are available in the file below.

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Working%20with%20Facts%20and%20Dimension%20Tables/star-schema.sql

Download the file by running the command below.

bash

```bash
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IB
```

Run

Step 4: Create the schema

Run the command below to create the schema in the under `billingDW` database.

bash

```bash
psql  -h postgres -U postgres -p 5432 billingDW < star-schema.sql
```

Run

You should see an ouput similar to the one below.

```
theia@theiadocker-rsannareddy:/home/project$ psql  -h localhost -U postgres -p 5432 billingDW < star-schema.sql
BEGIN
CREATE TABLE
CREATE TABLE
CREATE TABLE
ALTER TABLE
ALTER TABLE
COMMIT
theia@theiadocker-rsannareddy:/home/project$
```

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# Practice exercises

In this practice exercise, you will analyze the below csv file, which contains data about the daily sales at different stores of an international fashion retailer.

| storeid | country | city | date | totalsales |
|---|---|---|---|---|
| 1 | Japan | Tokyo | 01 February 2020 | 20300.50 |
| 2 | UK | London | 01 February 2020 | 34000.20 |
| 3 | USA | New York | 01 February 2020 | 28900.00 |
| 4 | USA | Chicago | 01 February 2020 | 27690.00 |
| 5 | France | Paris | 01 February 2020 | 12090.00 |

1. Problem:

> *Design the schema for the dimension table DimStore.*

▶ Click here for Hint

▶ Click here for Solution

2. Problem:

> *Design the schema for the dimension table DimDate.*

▶ Click here for Hint

▶ Click here for Solution

3. Problem:

> *Design the schema for the fact table FactSales.*

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▶ Click here for Hint

▶ Click here for Solution

**Congratulations! You have successfully finished this lab.**

## Authors

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# Exercise 5: Create the schema on the data warehouse

Step 1: Start the postgresql server.

Start the PostgreSQL server by clicking the command below:

Open and Start PostgreSQL in IDE

Step 2: Create the database on the data warehouse.

Using the createdb command of the PostgreSQL server, we can directly create the database from the terminal.

Firstly, run the command below to set your PostgreSQL password for authentication. Replace with your actual PostgreSQL password, and then execute

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```bash
export PGPASSWORD=<your_password>
```

Run

Now, run the command below to create a database named billingDW.

```bash
createdb -h postgres -U postgres -p 5432 billingDW
```

Run

In the above command

- -h mentions that the database server is accessible using the hostname "postgres"
- -U mentions that we are using the user name postgres to log into the database
- -p mentions that the database server is running on port number 5432

You should see an output like this.

```
theia@theiadocker-rsannareddy:/home/project$ createdb -h localhost -U postgres -p 5432 billingDW
theia@theiadocker-rsannareddy:/home/project$
```

Step 3: Download the schema .sql file.

The commands to create the schema are available in the file below.

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Working%20with%20Facts%20and%20Dimension%20Tables/star-schema.sql

Download the file by running the command below.

```bash
wget https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IB
```

Run

Step 4: Create the schema

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Run the command below to create the schema in the under `billingDW` database.

```bash
psql  -h postgres -U postgres -p 5432 billingDW < star-schema.sql
```

Run

You should see an ouput similar to the one below.

```
theia@theiadocker-rsannareddy:/home/project$ psql  -h localhost -U postgres -p 5432 billingDW < star-schema.sql
BEGIN
CREATE TABLE
CREATE TABLE
CREATE TABLE
ALTER TABLE
ALTER TABLE
COMMIT
theia@theiadocker-rsannareddy:/home/project$
```

← Exercise 4: Create a star …

Practice exercises →

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